

MYSQL

ASSIGNMENT 2- QUERYING DATA

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INTRODUCTION

Continuing from the previous assignment (database & table creation with constraints), I've used the provided DML insert codes to populate the tables from the given Employee Data. I've executed the queries in this assignment after inserting the data.

DATABASE USED:

```
USE employee;
```

TABLES USED:

- `SELECT * FROM EMPLOYEES;`
- `SELECT * FROM LOCATION;`
- `SELECT * FROM DEPARTMENTS;`

INSERTION OF PROVIDED EMPLOYEE_DATA:

- `INSERT INTO DEPARTMENTS (DEPARTMENT_ID,DEPARTMENT_NAME) VALUES
(1, 'Software Development'),
(2, 'Marketing'),`

- `INSERT INTO LOCATION(LOCATION) VALUES
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');`

- `INSERT INTO EMPLOYEES (EMPLOYEE_ID, EMPLOYEE_NAME, GENDER, AGE, HIRE_DATE, DESIGNATION, DEPARTMENT_ID, LOCATION_ID, SALARY) VALUES
(5001, 'Vikash Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
(5002, 'Ravish Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000).`

QUERIES:

- **Distinct Values:**

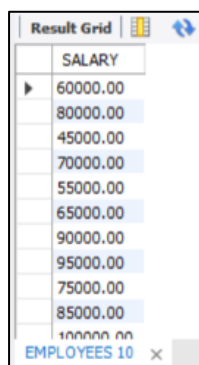
The query here retrieved and displayed the distinct salaries from the Employees table.

DISTINCT removes duplicate values from the result.

QUERY:

```
SELECT DISTINCT SALARY FROM EMPLOYEES;
```

OUTPUT:



The screenshot shows a 'Result Grid' window with a single column titled 'SALARY'. It contains 10 rows of distinct salary values: 60000.00, 80000.00, 45000.00, 70000.00, 55000.00, 65000.00, 90000.00, 95000.00, 75000.00, and 85000.00. The window title is 'EMPLOYEES 10'.

SALARY
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00

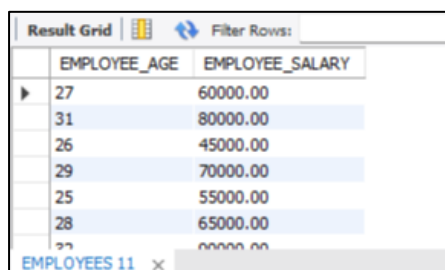
- **Alias (AS):**

Provided aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively which in result rename columns temporarily without changing the actual column name.

QUERY:

```
SELECT AGE AS EMPLOYEE_AGE,  
       SALARY AS EMPLOYEE_SALARY FROM EMPLOYEES;
```

OUTPUT:



The screenshot shows a 'Result Grid' window with two columns: 'EMPLOYEE_AGE' and 'EMPLOYEE_SALARY'. It contains 11 rows of data. The window title is 'EMPLOYEES 11'.

EMPLOYEE_AGE	EMPLOYEE_SALARY
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
27	85000.00

- Where Clause & Operators: To filter records using conditions.
- Retrieved employees with a salary greater than ₹50000 and hired before 2016-01-01.

QUERY:

```
SELECT * FROM EMPLOYEES
WHERE SALARY > 50000 AND HIRE_DATE < '2016-01-01';
```

OUTPUT:

EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID
Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4
Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1
Kiara Malhotra	F	29	2015-07-05	NULL	8	3
Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1
Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Updated and displayed the employee whose designation is missing and fill it with "Data Scientist".
-

QUERY:

```
UPDATE EMPLOYEES SET DESIGNATION = 'Data scientist'
WHERE DESIGNATION IS NULL;
```

OUTPUT:

5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1

5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2
5004	Kiara Malhotra	F	29	2015-07-05	Data scientist	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1

Sorting and Grouping Data:

- **ORDER BY:**

Displayed the employees sorted by department ID in ascending order and salary in descending order.

QUERY:

```
SELECT * FROM EMPLOYEES
ORDER BY DEPARTMENT_ID ASC,
SALARY DESC;
```

OUTPUT:

DEPARTMENT_ID	LOCATION_ID	SALARY
1	1	90000.00
2	3	90000.00
2	2	70000.00
2	1	60000.00
3	4	100000.00
3	2	75000.00
3	4	60000.00
4	4	100000.00

- **LIMIT:**

Displayed the first 5 employees hired in the year 2018.

QUERY:

```
SELECT * FROM EMPLOYEES
WHERE HIRE_DATE >= '2018-01-01'
AND HIRE_DATE <= '2019-01-01'
ORDER BY HIRE_DATE ASC
LIMIT 5;
```

OUTPUT:

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATI
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2

- Aggregate Functions:

- *SUM:*

Calculated the sum of all salaries in the Finance department.

QUERY:

```
SELECT SUM(SALARY) FROM EMPLOYEES e
JOIN DEPARTMENTS d
ON e.DEPARTMENT_ID=d.DEPARTMENT_ID
WHERE DEPARTMENT_NAME='FINANCE';
```

OUTPUT:

Result Grid	Filter Rows:
SUM(SALARY)	
170000.00	

- *MIN:*

Displayed the minimum age among all employees.

QUERY:

```
SELECT MIN(AGE) FROM EMPLOYEES;
```

OUTPUT:

Result Grid	Filter Rows:
MIN(AGE)	
25	

- GROUP BY:

- Listed the maximum salary for each location.

QUERY:

```
SELECT MAX(SALARY), l.LOCATION FROM EMPLOYEES e
JOIN LOCATION l
ON e.LOCATION_ID=l.LOCATION_ID
GROUP BY l.LOCATION_ID,l.LOCATION;
```

OUTPUT:

Result Grid	Filter Rows:
MAX(SALARY)	LOCATION
95000.00	Bangalore
95000.00	Chennai
90000.00	Hyderabad
100000.00	Pune

- Calculated the average salary for each designation containing the word 'Analyst'.

QUERY:

```
SELECT AVG(SALARY),DESIGNATION FROM EMPLOYEES
WHERE DESIGNATION LIKE '%Analyst%'
GROUP BY DESIGNATION;
```

OUTPUT:

Result Grid	Filter Rows:
AVG(SALARY)	DESIGNATION
60000.000000	Data Analyst
60000.000000	Marketing Analyst
60000.000000	Supply Chain Analyst
85000.000000	Financial Analyst
85000.000000	Finance Analyst

- HAVING:
- Displayed the departments with less than 3 employees.

QUERY:

```
SELECT DEPARTMENT_ID,COUNT(*) FROM EMPLOYEES
GROUP BY DEPARTMENT_ID HAVING COUNT(*)<3;
```

OUTPUT:

Result Grid	Filter Rows:
DEPARTMENT_ID	COUNT(*)
1	1
5	2
6	2
7	2
9	1
11	2
13	1

- Displayed the locations with female employees whose average age is below 30.

QUERY:

```
SELECT AVG(AGE), l.LOCATION FROM EMPLOYEES e
INNER JOIN LOCATION l
ON e.LOCATION_ID=l.LOCATION_ID
WHERE GENDER='F'
GROUP BY l.LOCATION
HAVING AVG(AGE)<30;
```

OUTPUT:

Result Grid	Filter Rows:
AVG(AGE)	LOCATION
27.3333	Bangalore
26.7500	Chennai
29.2000	Pune

Joins:

- **Inner Join:**

Listed employee names, their designations, and department names where employees are assigned to a department.

QUERY:

```
SELECT e.EMPLOYEE_NAME,e.DESIGNATION,d.DEPARTMENT_NAME FROM EMPLOYEES e
INNER JOIN DEPARTMENTS d
ON e.DEPARTMENT_ID=d.DEPARTMENT_ID;
```

OUTPUT:

EMPLOYEE_NAME	DESIGNATION	DEPARTMENT_NAME
Anvi Chaudhary	Business Development Executive	Business Development
Kaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support

- **Left Join:**

Listed all departments along with the total number of employees in each department, including departments with no employees.

QUERY:

```
SELECT d.DEPARTMENT_NAME, COUNT(EMPLOYEE_ID) FROM DEPARTMENTS d
LEFT JOIN EMPLOYEES e
ON d.DEPARTMENT_ID=e.DEPARTMENT_ID
GROUP BY d.DEPARTMENT_ID,d.DEPARTMENT_NAME;
```

OUTPUT:

DEPARTMENT_NAME	COUNT(EMPLOYEE_ID)
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4

- **Right Join:**

Displayed all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

QUERY:

```
SELECT l.location, e.employee_name FROM employees e
RIGHT JOIN location l
ON e.location_id = l.location_id;
```

OUTPUT:

Result Grid   Filter Rows		
	location	employee_name
▶	Bangal	Bangalore Iya Iyer
	Bangalore	Dhiruv Shetty
	Bangalore	Myra Verma
	Bangalore	Vihaan Mohan
	Bangalore	Kabir Nair
	Bangalore	Anika Paul
	Bangalore	Vivaan Singh
	Bangalore	Arjun Kumar
	Chennai	Reyansh Singh
	Chennai	Anvi Chaudhary
	Chennai	Saanvi Patel
	Chennai	Ishaan Kumar
	Chennai	Aarav Nair
	Chennai	Aryan Shetty
	Chennai	Anaya Kapoor
	Chennai	Sara T...

GLIMPSES OF QUERIES:

USE employee;

SELECT * FROM EMPLOYEES;

SELECT * FROM LOCATION;

SELECT * FROM DEPARTMENTS;

INSERT INTO DEPARTMENTS (DEPARTMENT_ID, DEPARTMENT_NAME) VALUES

(1, 'Software Development'),

(2, 'Marketing'),

(3, 'Data Science'),

(4, 'Human Resources'),

(5, 'Product Management'),

(6, 'Content Creation'),

(7, 'Finance'),

(8, 'Design'),

(9, 'Research and Development'),

(10, 'Customer Support'),

(11, 'Business Development'),

(12, 'IT'),

(13, 'Operations');

INSERT INTO LOCATION (LOCATION) VALUES

('Chennai'),

('Bangalore'),

('Hyderabad'),

('Pune');

```

INSERT INTO EMPLOYEES (EMPLOYEE_ID, EMPLOYEE_NAME, GENDER, AGE,
HIRE_DATE, DESIGNATION, DEPARTMENT_ID, LOCATION_ID, SALARY) VALUES

(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1,
55000),
(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),
(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),
(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),
(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),
(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),
(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),
(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),
(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),
(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),
(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),
(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),
(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),
(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),
(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),
(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),
(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),
(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),

```

(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),
(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),
(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),
(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),
(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);

SELECT DISTINCT SALARY FROM EMPLOYEES;

SELECT AGE AS EMPLOYEE_AGE,
SALARY AS EMPLOYEE_SALARY FROM EMPLOYEES;

SELECT * FROM EMPLOYEES
WHERE SALARY>50000 AND HIRE_DATE< '2016-01-01';

UPDATE EMPLOYEES SET DESIGNATION ='Data scientist'
WHERE EMPLOYEE_ID=5004;

SELECT * FROM EMPLOYEES
ORDER BY DEPARTMENT_ID ASC,
SALARY DESC;

SELECT * FROM EMPLOYEES
WHERE HIRE_DATE>='2018-01-01'
AND HIRE_DATE<='2019-01-01'
ORDER BY HIRE_DATE ASC
LIMIT 5;

```
SELECT SUM(SALARY) FROM EMPLOYEES e
JOIN DEPARTMENTS d
ON e.DEPARTMENT_ID= d.DEPARTMENT_ID
WHERE DEPARTMENT_NAME='FINANCE';
```

```
SELECT MIN(AGE) FROM EMPLOYEES;
```

```
SELECT MAX(SALARY), l.LOCATION FROM EMPLOYEES e
JOIN LOCATION l
ON e.LOCATION_ID=l.LOCATION_ID
GROUP BY l.LOCATION_ID,l.LOCATION;
```

```
SELECT AVG(SALARY), DESIGNATION FROM EMPLOYEES
WHERE DESIGNATION LIKE '%Analyst%'
GROUP BY DESIGNATION;
```

```
SELECT DEPARTMENT_ID, COUNT(*) FROM EMPLOYEES
GROUP BY DEPARTMENT_ID HAVING COUNT(*)<3;
```

```
SELECT AVG(AGE), l.LOCATION FROM EMPLOYEES e
INNER JOIN LOCATION l
ON e.LOCATION_ID=l.LOCATION_ID
WHERE GENDER='F'
GROUP BY l.LOCATION
HAVING AVG(AGE)<30;
```

```
SELECT e.EMPLOYEE_NAME,e.DESIGNATION,d.DEPARTMENT_NAME FROM  
EMPLOYEES e
```

```
INNER JOIN DEPARTMENTS d
```

```
ON e.DEPARTMENT_ID=d.DEPARTMENT_ID;
```

```
SELECT d.DEPARTMENT_NAME, COUNT(EMPLOYEE_ID) FROM DEPARTMENTS d
```

```
LEFT JOIN EMPLOYEES e
```

```
ON d.DEPARTMENT_ID=e.DEPARTMENT_ID
```

```
GROUP BY d.DEPARTMENT_ID,d.DEPARTMENT_NAME;
```

```
SELECT l.location, e.employee_name FROM employees e
```

```
RIGHT JOIN location l
```

```
ON e.location_id = l.location_id;
```

CONCLUSION

This assignment managed to attempt analyse the EMPLOYEE_DATA using MySQL. Various SQL queries such as data retrieval, filtering, sorting, aggregate functions, joins, grouping were applied. The assignment helped to understand how tables can be connected and retrieved for the purpose of further data updation.